Daily Temperature Forecasting with Seasonal Regression and ARMA Errors

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Abstract

This study examines the predictive contribution of annual seasonality and short-term residual dependence to daily mean temperature forecasting at Pullman/Moscow Regional Airport. Using 2,191 NOAA GSOD observations from 2019–2024, seasonal regression with ARMA errors is compared with persistence, independent-error seasonal regression, and nonseasonal ARIMA. Chronological validation selects three harmonic pairs with ARMA(1,1) errors. Following refitting through 2023, a rolling-origin evaluation with fixed parameters in 2024 yields RMSEs of 2.405 °C and 4.019 °C at one- and seven-day horizons. The seven-day improvement over seasonal regression is 0.015 °C. Empirical coverage of nominal 95% prediction intervals is 93.7% and 95.6%, respectively, with pronounced winter undercoverage. The findings support residual dependence modeling at the one-day horizon, while indicating limited incremental seven-day accuracy and deficiencies in constant-variance Gaussian uncertainty estimates.

1 Introduction

Daily temperature exhibits annual variation together with persistent departures from its seasonal mean. Distinguishing these components is relevant to interpretable forecasts for weather-sensitive planning. This study evaluates whether their joint representation improves one- and seven-day predictions at an eastern Washington station. The contribution is a controlled empirical comparison that isolates seasonal structure and residual dependence, while assessing predictive uncertainty under chronological data separation.

Related work

Dynamic harmonic regression combines Fourier predictors with ARMA errors, providing a parsimonious representation of long seasonal periods [1]. The harmonic count controls the flexibility of the seasonal mean, whereas the error process represents dependence unexplained by calendar terms. Its fixed seasonal shape constrains adaptation to atypical temperature departures, motivating horizon-specific evaluation.

Hyndman and Khandakar [2] develop automatic forecasting procedures that include stepwise ARIMA order selection. Their work provides a basis for systematic specification search. Here, a bounded exhaustive grid and chronological validation criterion replace automatic selection, permitting component attribution under a common forecast protocol.

Taylor and Letham [3] propose a modular additive approach with interpretable trend, seasonal, and holiday components. The present analysis emphasizes the incremental contribution of explicit ARMA residual dependence. Prophet is outside the empirical comparison; conclusions are restricted to the evaluated baselines.

Preprint.

2 Data and preprocessing

2.1 Data source and station selection

The analysis uses NOAA's Global Surface Summary of the Day (GSOD) archive [4] for January 1, 2019 through December 31, 2024. GSOD aggregates subdaily observations into daily summaries. The response variable is the reported daily mean temperature, rather than an estimate derived from daily extrema. Daily aggregation follows Greenwich Mean Time, so all observations and forecast targets are indexed by UTC date.

The selected station is Pullman/Moscow Regional Airport (USAF 727857, WBAN 94129; GSOD identifier 72785794129; ICAO KPUW). The inventory locates the station at 46.744 °N, 117.116 °W and 775.7 m elevation. Spokane International Airport (72785024157) was assessed as an alternative. Pullman's 99.954% temperature coverage supports its selection for the analysis. Although Spokane has complete coverage, Pullman provides the geographically preferred series with adequate availability. Station selection is independent of test-period forecast accuracy. Table 1 summarizes the chronological partitions.

Table 1: Chronological partitions at the Pullman station.

Partition	Calendar days	Observed	Dates
Training	1,461	1,460	2019–2022
Validation	365	365	2023
Test	366	366	2024

2.2 Quality control and missing observations

Station-year CSV files are retained in their original form, with source URLs and SHA-256 checksums recorded for traceability. Preprocessing verifies station identity, examines duplicate dates, and distinguishes exact duplicates from conflicting records. No duplicate dates occur in the selected files. The temperature missing-value sentinel, 9999.9, is absent from the retrieved observations. The `TEMP_ATTRIBUTES` field records the number of observations contributing to a daily mean; it is not an independent quality flag. All supplied means have positive observation counts.

Temperatures are converted to Celsius using $C = (F - 32)5/9$ and reindexed to the complete daily calendar. This retains leap days and the missing observation on December 31, 2021. That observation remains unfilled. State-space estimation propagates the latent state across the missing measurement, while independent seasonal regression is estimated from observed temperatures. Unobserved targets are excluded from evaluation by construction. In this dataset, both validation and test targets are complete.

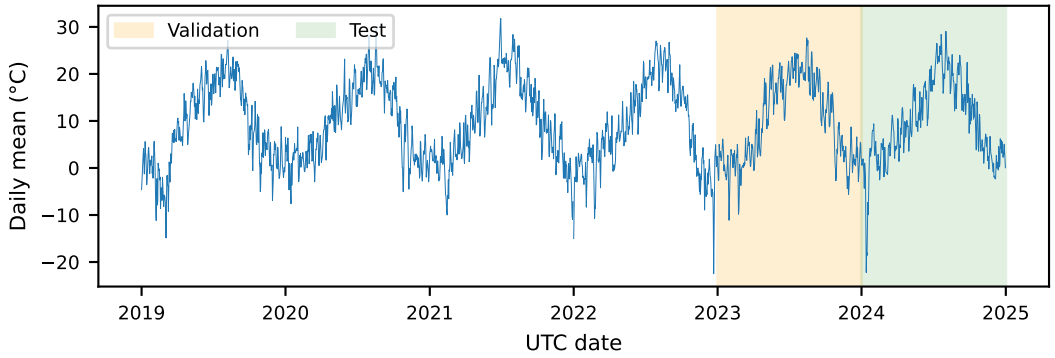


Figure 1: Daily temperatures and the fixed evaluation partitions. The strong annual cycle motivates known future calendar predictors.

3 Methodology

3.1 Seasonal regression with ARMA/ARIMA errors

Let t be elapsed calendar days since January 1, 2019. For a fixed annual period $P = 365.25$, the model is

$$y_t = s_t + u_t, \tag{1}$$

$$s_t = \beta_0 + \sum_{k=1}^{K} \left[a_k \sin\left(\frac{2\pi k t}{P}\right) + b_k \cos\left(\frac{2\pi k t}{P}\right) \right]. \tag{2}$$

The kth sine–cosine pair represents a harmonic with period P/k. Increasing K permits a more flexible seasonal profile while retaining a low-dimensional parameterization. The fractional period approximates annual seasonality rather than exact calendar anniversaries. A common time origin preserves phase across data partitions and missing observations. The seasonal coefficients are constant within each evaluation period.

The residual process satisfies

$$\phi(B)(1-B)^d u_t = \theta(B)\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

where B is the lag operator, $\phi(B) = 1 - \sum_{j=1}^{p} \phi_j B^j$, and $\theta(B) = 1 + \sum_{j=1}^{q} \theta_j B^j$. When $d = 0$, the specification reduces to ARMA(p, q) errors. If diagnostic evidence supports $d = 1$, the regression intercept is omitted because differencing eliminates its contribution. This formulation follows dynamic harmonic regression [1]. The training diagnostics reported below support $d = 0$ for the combined models evaluated here.

Regression coefficients, ARMA parameters, and innovation variance are estimated jointly by maximum likelihood using the SARIMAX state-space implementation in `statsmodels` [5]. An intercept is included in the Fourier design matrix, with no additional SARIMAX trend. Parameter estimation enforces stationarity and invertibility. Conditional on the information available through origin t, the forecast decomposes as

$$\widehat{y}_{t+h|t} = \widehat{s}_{t+h} + \widehat{u}_{t+h|t}. \tag{4}$$

The seasonal term is determined by the future date, whereas the residual expectation is obtained recursively from the filtered state. Prediction intervals are derived from the conditional state-space forecast distribution for future observations, treating estimated parameters as fixed.

3.2 Baseline models and component attribution

Persistence uses $\widehat{y}_{t+h|t} = y_t$ at both horizons, providing a reference based solely on the most recent observed temperature. All evaluated origins have an observed value of y_t.

Seasonal regression retains Equation 2 with independent Gaussian errors, implemented as ARIMA$(0, 0, 0)$ errors. This baseline isolates the predictive contribution of deterministic annual structure.

Nonseasonal ARIMA omits Fourier predictors and searches $p, q \in \{0, 1, 2, 3\}$ with $d \in \{0, 1\}$. A constant is included for $d = 0$ and omitted for $d = 1$; differenced candidates have no drift. This family represents serial dependence without explicit annual regressors.

The **combined model** includes both Fourier predictors and residual dynamics. Its candidate set contains ARMA$(0, 0)$, nesting the independent-error seasonal baseline. Consequently, validation can select a model without residual dependence if that restriction improves predictive accuracy.

3.3 Parameter interpretation

The selected $K = 3$ model refitted through 2023 has seasonal intercept 9.254 °C, $\widehat{\phi}_1 = 0.621$, $\widehat{\theta}_1 = 0.358$, and $\widehat{\sigma}^2 = 5.623\,^{\circ}\mathrm{C}^2$. For ARMA(1,1), residual mean forecasts beyond the first step decay by successive powers of $\widehat{\phi}_1$. This decay implies progressively smaller contributions from recent residual information at longer horizons. Complete coefficient estimates are retained with the reproducibility artifacts and remain fixed throughout testing.

4 Experimental design

4.1 Chronological validation and model selection

Candidate parameters are estimated from 2019–2022. Validation proceeds through 2023 using daily forecast origins, with fixed parameters and sequential state updates. The first origin is December 31, 2022. Eligibility requires the target date to fall within 2023, yielding 365 one-day and 359 seven-day predictions. The first seven-day target is January 7. Every model is evaluated on the same origin–target pairs, ensuring comparability across specifications.

We search $K \in \{1, 2, 3\}$ and $p, q \in \{0, 1, 2, 3\}$ for combined models. Training ordinary seasonal-regression residuals are checked with augmented Dickey–Fuller and KPSS tests. We consider $d = 1$ for a harmonic count only if ADF does not reject a unit root at 5% and KPSS rejects level stationarity at 5%. No harmonic count satisfies both conditions. This results in 48 combined candidates, three independent-error seasonal candidates, and 32 nonseasonal ARIMA candidates: 83 fits in total. All converge. These stationarity checks are preliminary diagnostics; their residual input drops the single missing training value and therefore closes one calendar gap. The likelihood fits and forecast calendars preserve that gap.

One specification is selected within each model family by minimizing the equal-weight mean of validation RMSE across the two horizons:

$$S(m) = \tfrac{1}{2} \left(\mathrm{RMSE}_1(m) + \mathrm{RMSE}_7(m) \right). \tag{5}$$

Both components are expressed in Celsius. This criterion selects a common specification across horizons and makes the weighting of short- and longer-horizon accuracy explicit.

We record AIC, convergence warnings, parameters, and residual checks, but AIC does not determine the selected family specifications. Nonconverged fits would be excluded. The optimizer uses up to 300 iterations with a Powell-assisted retry if needed, followed by another likelihood optimization.

4.2 Held-out evaluation and information constraints

Following model selection, each retained specification is refitted on 2019–2023. Orders are recorded before evaluating 2024, and all parameter estimates remain fixed throughout the test period. At origin t, the model uses only observations available through t to generate predictions at $t + 1$ and $t+7$. The next observation is incorporated through a filtering update only after these predictions have been recorded. Repeating this operation produces daily rolling-origin forecasts without parameter refitting.

The first test origin is December 31, 2023. Target-date restrictions yield 366 one-day and 360 seven-day predictions. In particular, observations between t and $t + 7$ cannot influence the forecast issued at t for the seven-day target. The information-availability assumption is that completed UTC-day summaries are available at the origin. Actual publication delays are outside the evaluation protocol.

4.3 Evaluation criteria and protocol verification

For errors $e_i = \widehat{y}_i - y_i$ over n eligible targets, we report $\mathrm{RMSE} = \sqrt{n^{-1} \sum_i e_i^2}$ and $\mathrm{MAE} = n^{-1} \sum_i |e_i|$ separately by horizon. For the combined model, coverage is $n^{-1} \sum_i \mathbf{1}\{L_i \le y_i \le U_i\}$ and mean width is $n^{-1} \sum_i (U_i - L_i)$ for nominal 95% intervals. Implementation checks verify horizon alignment and common target sets. A perturbation check confirms that changing future observations leaves previously issued forecasts unchanged.

5 Validation results and selected specifications

Table 2 presents validation performance for the selected specification in each family. Seasonal regression retains three Fourier pairs, the nonseasonal baseline retains ARIMA$(3, 0, 3)$, and the combined model retains three harmonic pairs with ARMA$(1, 1)$ errors. The latter comprises seven seasonal coefficients, two dependence parameters, and an innovation variance. Selection therefore favors a lower-order residual process within the specified search space.

Table 2: 2023 validation metrics. Each model is fitted on 2019–2022. One-day metrics use 365 targets and seven-day metrics use 359.

	One day		Seven days	
Model	RMSE	MAE	RMSE	MAE
Persistence	2.441	1.881	5.248	4.039
Seasonal ($K = 3$)	3.661	2.896	**3.672**	**2.903**
ARIMA$(3, 0, 3)$	2.259	1.763	4.457	3.429
Combined ($K = 3$, ARMA$(1, 1)$)	**2.169**	**1.703**	3.677	2.906

The one-day results indicate that recent observations contain information beyond the deterministic calendar mean. Persistence outperforms seasonal regression, and nonseasonal ARIMA improves further. The combined model attains the lowest one-day RMSE, 2.169 °C, suggesting an additional contribution from explicit seasonal predictors. At seven days, seasonal regression yields a marginally lower RMSE than the combined model, 3.672 versus 3.677 °C. Both substantially outperform persistence and nonseasonal ARIMA at this horizon.

These comparisons indicate a horizon-dependent contribution of residual dynamics. Their incremental predictive value is evident at one day but negligible at seven days in validation. The cross-horizon selection criterion identifies a common combined specification without implying improvement over the seasonal baseline at every horizon.

Training ADF p-values for the three seasonal residual series range from about 1.2×10^{-14} to 4.5×10^{-12}. KPSS reports 0.1 for each; this is a lookup-table bound, not an exact p-value. These checks support the decision to omit differencing in the combined-model search.

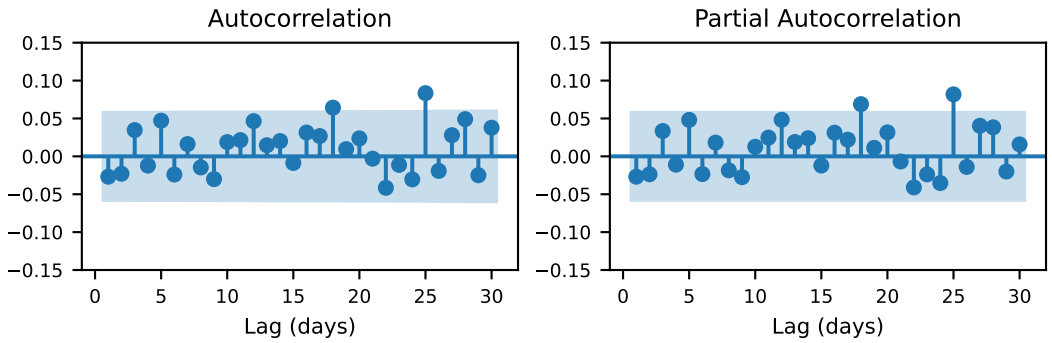


Figure 2: ACF and PACF of the selected combined model's standardized training innovations. The longest continuous observed segment is used after a 30-day burn-in. Shading is the plotting routine's nominal pointwise band.

For this continuous segment, Ljung–Box tests yield p-values of 0.517 and 0.169 at lags 14 and 30, respectively, with degrees of freedom adjusted by $p + q$. Figure 2 shows limited remaining autocorrelation. The plotted bands are pointwise rather than simultaneous bounds across lags. Failure to reject residual autocorrelation at these lags provides diagnostic support for the conditional mean specification, but does not establish normality or homoscedasticity.

6 Held-out point forecast performance

Table 3 reports held-out performance after refitting the selected models on 2019–2023. The combined model attains the lowest observed RMSE and MAE at both horizons. Its one-day RMSE is 2.405 °C, compared with 2.578 for persistence, 4.003 for seasonal regression, and 2.486 for nonseasonal ARIMA. The one-day ordering is consistent with validation and supports the joint representation of annual structure and residual dependence.

Table 3: 2024 held-out point forecast metrics in Celsius. The common sample has 366 one-day and 360 seven-day targets. Bold indicates the minimum within each column; it does not indicate a significance test.

	One day		Seven days	
Model	RMSE	MAE	RMSE	MAE
Persistence	2.578	1.978	5.556	4.371
Seasonal	4.003	2.995	4.034	3.028
Nonseasonal ARIMA	2.486	1.932	4.903	3.787
Combined	**2.405**	**1.815**	**4.019**	**3.022**

Relative to persistence, the combined specification reduces test RMSE by approximately 6.7% at one day and 27.7% at seven days. Reductions relative to nonseasonal ARIMA are approximately 3.2% and 18.0%, respectively. However, the seven-day improvement over persistence principally reflects the contribution of annual structure: seasonal regression alone achieves nearly the same accuracy as the combined model.

At seven days, the combined and seasonal models have RMSEs of 4.019 °C and 4.034 °C. The difference using unrounded scores is approximately 0.015 °C, or 0.37% of the seasonal baseline's RMSE. The corresponding MAE difference is approximately 0.005 °C. This small advantage, together with the reversed ordering in validation, limits the evidence for a material seven-day benefit from residual dependence modeling.

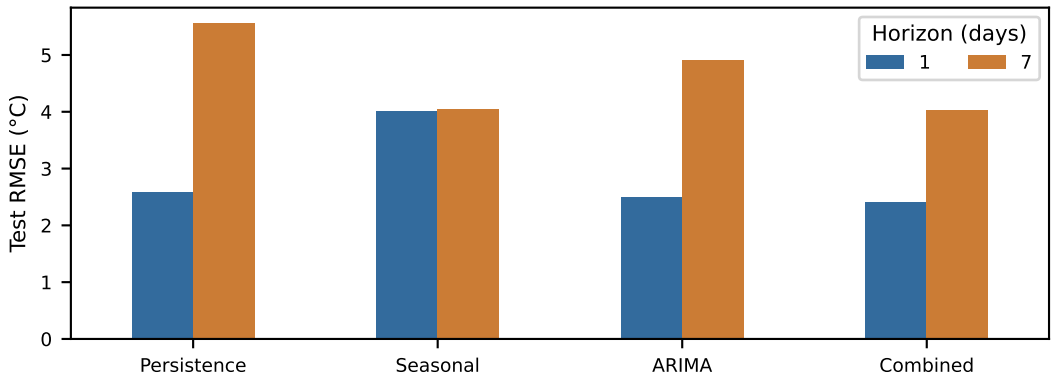


Figure 3: Test RMSE by model and horizon. Explicit annual seasonality is particularly valuable for seven-day forecasts; residual dynamics substantially improve on calendar-only regression at one day.

All comparisons use identical target dates, eliminating target-subset composition as an explanation for the observed differences. Nevertheless, these estimates describe one station and one test year. No statistical significance tests are reported, and overlapping seven-day errors are not assumed independent. The results therefore support conditional conclusions about this evaluation setting rather than general model dominance.

7 Prediction intervals and residual diagnostics

Table 4: Combined-model nominal 95% prediction intervals in the 2024 test. Width is measured in Celsius. Coverage is empirical over observed targets.

Horizon	Targets	Coverage	Mean width
One day	366	93.7%	9.295
Seven days	360	95.6%	14.856

Mean interval width increases with horizon, from 9.295 °C at one day to 14.856 °C at seven days, reflecting increased forecast uncertainty. Seven-day aggregate coverage is close to the nominal level, whereas one-day coverage is 93.7%. Point forecast improvement consequently does not imply nominal empirical coverage. Joint consideration of coverage and width is necessary because wider intervals may improve coverage at the expense of precision.

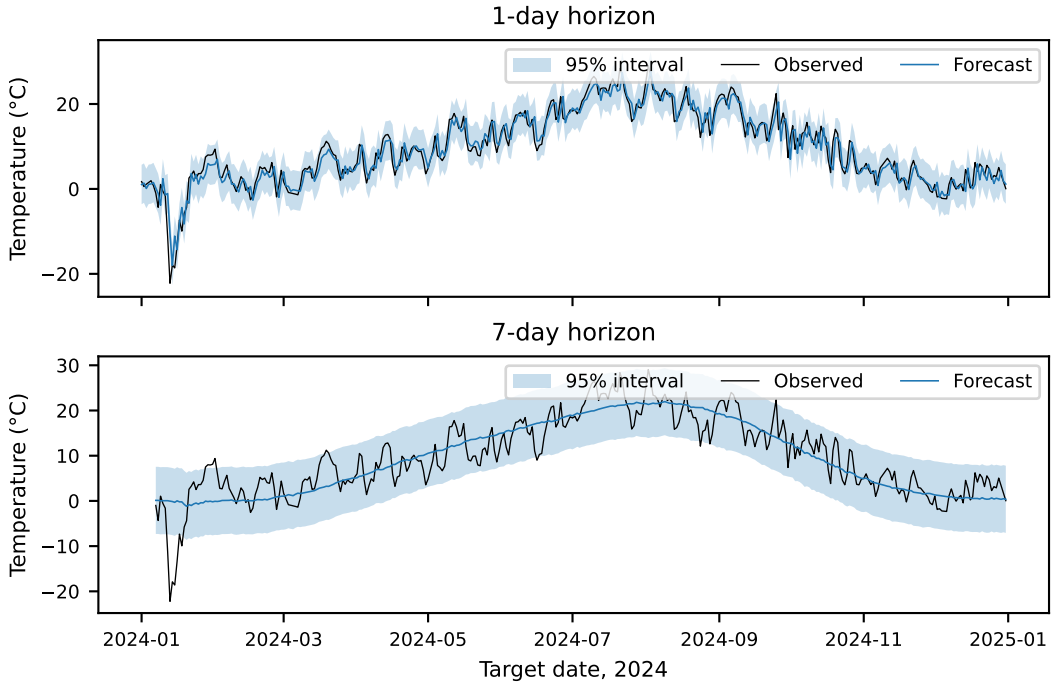


Figure 4: Rolling test predictions and conditional 95% intervals. Each plotted forecast is issued at its own origin; these curves are not a single forecast issued at the start of 2024. The seven-day forecast closely tracks the seasonal mean and misses the sharp January cold departure.

After refitting, the combined model yields Ljung–Box p-values of 0.459 and 0.158 at lags 14 and 30. Diagnostics are computed on the longest uninterrupted observed innovation segment, comprising 1,065 days after accounting for burn-in and the missing observation. In contrast, the Jarque–Bera test rejects normality ($p \approx 5.1 \times 10^{-13}$), and the ARCH LM test with 14 lags returns $p = 0.025$, indicating evidence of conditional heteroscedasticity. An adequate representation of serial mean dependence therefore does not ensure adequacy of the innovation distribution.

The Gaussian intervals condition on estimated coefficients and exclude parameter uncertainty. Their reported performance evaluates the original specification without recalibration against held-out outcomes. Alternative distributional assumptions or variance dynamics require a separate validation exercise before a further test evaluation.

8 Discussion and limitations

8.1 Seasonal heterogeneity in forecast performance

We group test targets into December–February (DJF), March–May (MAM), June–August (JJA), and September–November (SON). DJF combines January and February 2024 with December 2024; it is a seasonal category, not a single contiguous winter episode. There are 91, 92, 92, and 91 one-day targets in these groups. The seven-day winter group has 85 targets because the first six January dates are not seven-day targets under our origin rule.

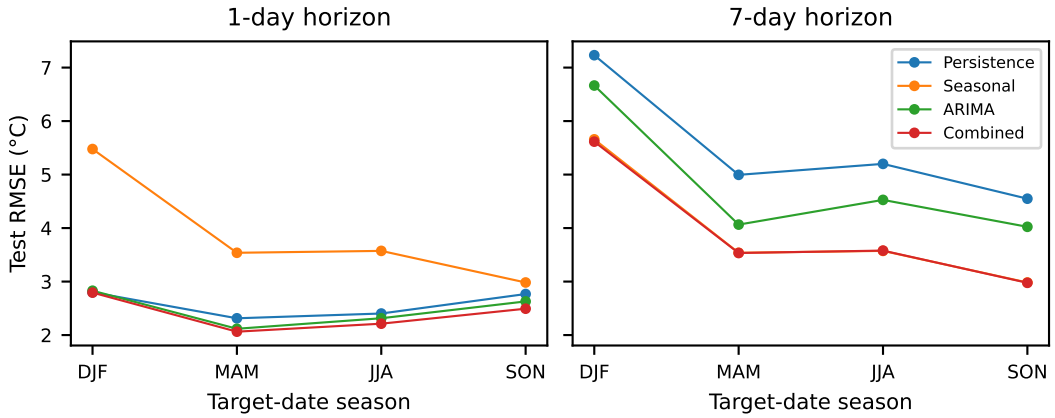


Figure 5: Target-season RMSE. Winter is the most difficult season at both horizons. Lines connect seasonal categories for readability and do not represent an interpolated time series or confidence intervals.

The combined model exhibits its largest errors in winter, with RMSEs of 2.794 °C and 5.616 °C at one and seven days, respectively. Seven-day RMSE declines to 3.534 in spring, 3.579 in summer, and 2.977 in autumn. Aggregate accuracy thus conceals substantial seasonal heterogeneity. The January cold departure visible in the forecast trajectories contributes to winter error, although this descriptive comparison does not identify the causes of individual forecast failures.

Winter interval coverage is 91.2% at one day and 87.1% at seven days. Seven-day coverage in spring, summer, and autumn is 98.9%, 96.7%, and 98.9%, respectively. Near-nominal annual coverage therefore combines winter undercoverage with relatively high coverage in other seasons. This heterogeneity limits the interpretation of aggregate calibration for weather-sensitive decisions, particularly during cold periods.

8.2 Scope and external validity

The predictor set contains only historical station observations and calendar terms. Consequently, the model has no direct access to atmospheric information that could anticipate changing weather regimes. The study establishes a transparent statistical benchmark, while leaving comparison with operational weather prediction systems outside its scope.

The six-year record provides four years for initial estimation and one year each for validation and testing. Geographic and temporal generalization remain limited by the single-station, single-test-year design. Coverage checks do not constitute independent verification of every source measurement. In addition, the fixed seasonal curve excludes secular trends and year-specific changes in seasonal shape.

The evaluation distinguishes sequential state adaptation from parameter re-estimation. It does not quantify the potential benefit of periodic refitting. The assumption of immediate availability of completed daily summaries further limits operational interpretation. Deployment would require alignment between forecast origins and actual data-release times, followed by prospective validation under those information constraints.

9 Reproducibility and conclusions

9.1 Computational reproducibility

The implementation comprises Python scripts for station auditing, preprocessing, model selection, rolling-origin evaluation, and result generation, together with an executed demonstration notebook. Original inputs are cached locally. Download manifests and SHA-256 checksums identify the source files, while reproduction hashes link the processed series, analysis code, evaluation protocol, and metric tables. Candidate forecasts and fitted parameter records permit independent inspection of the reported comparisons. The recorded environment uses Python 3.13.12, NumPy 2.4.3, pandas 3.0.1, statsmodels 0.14.6, and Matplotlib 3.10.8.

The notebook reconstructs a forecast from fixed coefficients using observations available only through a specified origin, and verifies its predictions and interval endpoints against the saved rolling evaluation. Additional protocol checks cover Fourier phase across leap days, exact horizon alignment, parameter preservation, and invariance of earlier forecasts to perturbations of future observations. All ten notebook code cells execute successfully. These checks establish implementation consistency rather than external predictive validity. Source code and artifacts accompany the study at https://github.com/lamanmirzayeva/Weather-Forecasting.

9.2 Conclusions and future research

The results provide evidence that annual Fourier regression and ARMA residual dynamics offer complementary information for one-day temperature forecasting at the studied station. The combined model achieves the lowest one-day RMSE in both chronological validation and held-out testing. At seven days, the incremental test improvement over seasonal regression is small and the validation ordering differs, indicating that annual structure accounts for most of the observed predictive benefit at that horizon.

Uncertainty evaluation yields a more qualified conclusion. Aggregate seven-day coverage is near nominal, but winter undercoverage, nonnormal innovations, and evidence of changing variance reveal limitations of the Gaussian constant-variance specification. Future research should assess alternative innovation distributions and variance models, additional stations, and prospective test years. Such extensions require renewed chronological validation rather than selection against the completed 2024 test.

References

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